

Image Quality Parameter Configuration User Guide

The upgraded rom can modify the filter value of the corresponding parameters when adding images through the AI service configuration.

Original AI service jsonData data structure:

```
{
    "work_mode": 0,
    "recognize_mode" : 1,
    "face_mode" : 1,
    "enable_qr_code" : false,
    "enable_track_box" : true,
    "liveness_thresh" : 0.98,
    "recognition_thresh" : 0.92,
    "verify_thresh" : 0.91,
    "recognition_interval" : 5,
    "report_interval" : 5,
    "max_face_cnt" : 4,
    "face_size" : 10,
    "enable_feature_export" : true,
    "enable_image_export" : true,
    "ae_mode" : 1,
    "ae_flicker_freq" : 0,
    "enable_safety_hat":false,
    "enable_mask_sign":false,
    "enable_extra_attribute":true
}
```

The updated AI service jsonData data structure: (the bolded part is the newly added quality configuration parameters)

```
{
    "work_mode": 0,
```

```
"recognize_mode" : 1,  
"face_mode" : 1,  
"enable_qr_code" : false,  
"enable_track_box" : true,  
"liveness_thresh" : 0.98,  
"recognition_thresh" : 0.92,  
"verify_thresh" : 0.91,  
"recognition_interval" : 5,  
"report_interval" : 5,  
"max_face_cnt" : 4,  
"face_size" : 10,  
"enable_feature_export" : true,  
"enable_image_export" : true,  
"ae_mode" : 1,  
"ae_flicker_freq" : 0,  
"enable_safety_hat":false,  
"enable_mask_sign":false,  
"enable_extra_attribute":true,  
"db_align_confidence": 0.4,  
"db_blur": 0.7,  
"db_detect_confidence": 0.3,  
"db_face_size_min": 100,  
"db_pitch_max": 30.001,  
"db_pitch_min": -30.001,  
"db_roll_max": 30.001,  
"db_roll_min": -30.001,  
"db_sum_quality": 0.3,  
"db_yaw_max": 30.001,  
"db_yaw_min": -30.001  
}
```

Description of new parameters:

Parameter name	Description	Default value	Configurable range
db_align_confidence	Minimum face alignment confidence. Alignment refers to aligning the detected face with the standard face, that is, mapping faces of different poses to a unified standard. Alignment confidence refers to the degree of coincidence between the key points of the face and the key points of the preset standard face. The higher the value, the higher the degree of coincidence.	0.4	[0,1]
db_blur	Minimum face blur score. The higher the score, the clearer the face.	0.7	[0,1]
db_detect_confidence	Minimum face detection confidence. The face detection confidence represents the credibility of the existence of a face in the picture. The higher the score, the higher the probability that there is a face in the picture.	0.3	[0,1]
db_face_size_min	Minimum face width.	100	[60,+ ∞)
db_pitch_max	Maximum pitch angle. The pitch angle refers to the angle of rotation around the X axis (that is,	30	[0,90]

	to the right side of the face), which is expressed as the angle of head pitch or head down. The absolute value of the maximum pitch angle indicates the maximum head down angle.		
db_pitch_min	Same as above. The absolute value of the minimum pitch angle indicates the maximum head pitch angle.	-30	[-90,0]
db_roll_max	Maximum roll angle. The roll angle refers to the rotation angle around the Z axis (that is, towards the front of the face), which is expressed as the rotation angle of the face photo seen from the front. The absolute value of the maximum roll angle represents the maximum angle of counterclockwise rotation.	30	[0,90]
db_roll_min	Same as above. The absolute value of the minimum roll angle represents the maximum angle of clockwise rotation.	-30	[-90,0]
db_yaw_max	Maximum yaw angle. Yaw angle refers to the angle of rotation around the Y-axis (that is, toward the face below), which is expressed as the angle of horizontal rotation of the head to the left or right. The	30	[0,90]

	absolute value of the maximum yaw angle represents the maximum angle to turn the head to the left.		
db_yaw_min	Same as above. The absolute value of the minimum yaw angle represents the maximum angle to turn the head to the right.	-30	[-90,0]
db_sum_quality	Minimum comprehensive quality score. The calculation method is: $\text{db_sum_quality} = \text{detect_confidence} * \text{align_confidence} * \text{blur} * \exp(-10 * \text{yaw} * \text{yaw}/8100)$	0.3	[0,1]